Arista Access Node HEB-ACC-43 Routing Protocol Flap Troubleshooting SOP

Node: HEB-ACC-43 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_2023

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on an Arista Access Node, specifically HEB-ACC-43 located at the Hebbal Edge site. The underlying cause of this failure domain is the dynamic nature of routing protocols, which can lead to temporary instability and fluctuations in the control plane. In Arista's EOS, BGP route churn occurs when there is a change in the route advertisements, leading to a series of BGP updates and withdrawals. Similarly, OSPF adjacency loss results from a change in the OSPF topology, causing the adjacency state to transition from Full to Down. Finally, ISIS LSP flooding occurs when there is a change in the ISIS link-state database, leading to a series of ISIS LSP updates and withdrawals. The severity of this alarm is Warning, indicating that the issue is not critical but requires attention to prevent potential network instability and impact on services.

Pre-requisite Checks

- Step: Verify site access and environmental conditions, Description: Ensure that the Hebbal Edge site is accessible and that the environmental conditions (temperature, humidity, power) are within the recommended operating range.

- Step: Check redundant link status, Description: Verify that the redundant link to the Hebbal Edge node is functioning correctly and is not contributing to the routing protocol flap.

- Step: Gather log files and configuration data, Description: Collect the log files and configuration data from the Hebbal Edge node to aid in analysis and troubleshooting.

Potential Root Causes

Cause	Probability
BGP route churn	High
OSPF adjacency loss	Medium
ISIS LSP flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP router identifier 10.10.10.10, local AS number 64512, Description: This command displays a summary of the BGP routing table, including the number of routes, peers, and updates. It can help identify any issues with the BGP routing protocol.
```

```
Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Interface, Description: This command displays the OSPF
neighbor table, including the neighbor ID, priority, state, and dead
time. It can help identify any issues with the OSPF adjacency.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LS ID/ Age), Description: This command displays the ISIS
link-state database, including the LSP sequence number, LS ID, and
age. It can help identify any issues with the ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Clear BGP updates, Command: clear bgp *, Description: This command clears all BGP updates, resetting the BGP routing table and peers. It can help resolve BGP route churn issues.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf neighbor *, Description: This command resets the OSPF adjacency, re-establishing the OSPF neighbor relationship. It can help resolve OSPF adjacency loss issues.

Step 3: Step: Flush ISIS LSP database, Command: clear isis database, Description: This command flushes the ISIS LSP database, resetting the ISIS link-state database. It can help resolve ISIS LSP flooding issues.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP router identifier
10.10.10.10, local AS number 64512, Description: This command displays
a summary of the BGP routing table, including the number of routes,
peers, and updates. It can be used to verify that the BGP routing
protocol is stable.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Interface, Description: This command displays the OSPF
neighbor table, including the neighbor ID, priority, state, and dead
time. It can be used to verify that the OSPF adjacency is established.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LS ID/ Age), Description: This command displays the ISIS
link-state database, including the LSP sequence number, LS ID, and
age. It can be used to verify that the ISIS LSP flooding is resolved.
```

Roll-back Procedures

- Step: Restore BGP configuration, Command: copy running-config startup-config, Description: This command restores the BGP configuration from the running configuration to the startup configuration, reverting any changes made during the troubleshooting process.

- Step: Reset OSPF adjacency, Command: clear ospf neighbor *, Description: This command resets the OSPF adjacency, re-establishing the OSPF neighbor relationship. It can be used to revert any changes made during the troubleshooting process.

- Step: Flush ISIS LSP database, Command: clear isis database, Description: This command flushes the ISIS LSP database, resetting the ISIS link-state database. It can be used to revert any changes made during the troubleshooting process.